



Programming Java

Applets

Incheon Paik



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An Overview of Applets

- ❑ Applet : A program that can be referenced by HTML source code of a Web page.
- ❑ Can be run on Web browser after having been downloaded.
- ❑ May be dangerous
- ❑ Security Problem



First Java Applet

```
import java.applet.Applet;
import java.awt.Graphics;
/*
<applet code="FirstApplet" width=200 height=200>
</applet>
*/

public class FirstApplet extends Applet {
    public void paint(Graphics g) {
        g.drawString("This is my first applet!", 20, 100);
    }
}
```

- ❑ Extends Applet
- ❑ Graphics by Abstract Window Toolkit (AWT)
- ❑ Run: after compile,
appletviewer FirstApplet.html
or
appletviewer FirstApplet.java

The Life Cycle of an Applet

```
import java.applet.Applet;
import java.awt.Graphics;
/*
<applet code="AppletLifecycle" width=300
height=50>
</applet>
*/

public class AppletLifecycle extends Applet {
    String str = "";

    public void init() {
        str += "init; ";
    }

    public void start() {
        str += "start; ";
    }

    public void stop() {
        str += "stop; ";
    }

    public void destroy() {
        System.out.println("destroy");
    }

    public void paint(Graphics g) {
        g.drawString(str, 10, 25);
    }
}
```



- `init()` : called only when the applet begins execution.
- `start()` : executed after `init()` method. Called by the applet viewer or Web browser.
- `stop()` : when applet viewer is minimized.
- `destroy()` : called by the applet viewer or Web browser before the applet is terminated.

The Graphics Class

Methods of Graphics Class

```
abstract void drawArc(int x, int y, int w, int h, int degrees0,
int degrees1)
abstract boolean drawImage(Image img, int x, int y, Image
Observer io)
abstract void drawLine(int x0, int y0, int x, int y1)
abstract void drawOval(int x, int y, int w, int h)
abstract void drawPolygon(int x[], int u[], int n)
abstract void drawPolyline(int x[], int y[], int n)
void drawRect(int x, int y, int w, int h)
abstract void drawString(String str, int x, int y)
abstract void fillArc(int x, int y, int w, int h, int degree0, int
degree1)
abstract void fillOval(int x, int y, int w, int h)
abstract void fillPolygon(int x[], int y[], int n)
void fillRect(int x, int y, int w, int h)
abstract Color getColor()
abstract Font getFont()
abstract FontMetrics getFontMetrics()
abstract void setColor(Color c)
abstract void setFont(Font f)
```

```
import java.applet.Applet;
import java.awt.Graphics;
/*
<applet code="DrawArc" width=200 height=200>
</applet>
*/

public class DrawArc extends Applet {

    public void paint(Graphics g) {
        g.drawArc(20, 20, 160, 160, 0, 135);
    }
}
```



Using Colors

Color Constructors

```
Color(int red, int green, int blue)
Color(int rgb)
Color(float r, float g, float b)
```

Method of Graphics Class

```
static int HSBtoRGB(float h, float s, float b)
static float[] RGBtoHSB(int r, int g, int b, float hsb[])
Color brighter()
Color darker()
static Color decode(String str) throws NumberFormatException
boolean equals(Object obj)
int getBlue()
int getGreen()
int getRGB()
int getRed()
```

<http://java.sun.com/j2se/1.4.2/docs/api/java/awt/Color.html>

```
import java.applet.Applet;
import java.awt.Color;
import java.awt.Graphics;
/*
<applet code="BlueString" width=300 height=100>
</applet>
*/
```

```
public class BlueString extends Applet {

    public void paint(Graphics g) {
        g.setColor(Color.blue);
        g.drawString("Blue String", 100, 50);
    }
}
```



Displaying Text

Font Constructor

```
Font(String name, int style, int ps)
```

setFont() Method

```
abstract void setFont(Font font)
```

FontMetrics Constructor

```
FontMetrics(Font font)
```

<http://java.sun.com/j2se/1.4.2/docs/api/java/awt/Font.html>

```
import java.applet.Applet;
import java.awt.*;
/*
<applet code="FontDemo" width=200 height=200>
</applet>
*/
```

```
public class FontDemo extends Applet {

    public void paint(Graphics g) {

        // Draw baseline
        int baseline = 100;
        g.setColor(Color.lightGray);
        g.drawLine(0, baseline, 200, baseline);

        // Draw String
        g.setFont(new Font("Serif", Font.BOLD, 36));
        g.setColor(Color.black);
        g.drawString("Wxyz", 5, baseline);
    }
}
```



Using Applet Dimensions

getSize() Method

Dimension getSize()

Dimension Constructors

Dimension()

Dimension(*Dimension* d)

Dimension(int w, int h)

```
import java.applet.*;
import java.awt.*;
/*
<applet code="Circle" width=200 height=200>
</applet>
*/
```

```
public class Circle extends Applet {
```

```
    public void paint(Graphics g) {
        Dimension d = getSize();
        int xc = d.width / 2;
        int yc = d.height / 2;
        int radius = (int)((d.width < d.height) ?
            0.4 * d.width : 0.4 * d.height);
        g.drawOval(xc - radius, yc - radius, 2 * radius, 2 *
            radius);
    }
}
```

<http://java.sun.com/j2se/1.5.0/docs/api/java/awt/Dimension.html>



Using Applets in a Web Page

Applet HTML Tag

```
<applet
[codebase=url]
[alt=text]
[name=appName]
width=wpixels
height=hpixels
[align=alignment]
[vspace=vspixels]
[hspace=hspace]
>
[<param name=pname1 value=value1>]
[<param name=pname2 value=value2>]
.....
[<param name=pnameN value=valueN>]
</applet>
```

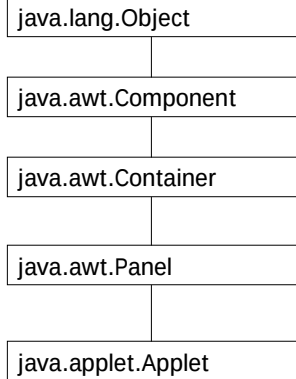
```
import java.applet.*;
import java.awt.*;
/*
<applet code="BackgroundForeground" width=200
height=200>
</applet>
*/
```

```
public class BackgroundForeground extends Applet {
```

```
    public void paint(Graphics g) {
        setBackground(Color.yellow);
        setForeground(Color.blue);
        g.drawLine(0, 0, 200, 200);
        g.fillRect(100, 40, 50, 50);
    }
}
```



The Applet Class



Applet and its superclasses

```
import java.applet.*;
import java.awt.*;
/*
<applet code="AppletParameters" width=300
height=300>
<param name="background" value="0xfffff">
<param name="foreground" value="0x000000">
<param name="message" value="Testing Applet
Parameters">
</applet>
*/

public class AppletParameters extends Applet {

    public void paint(Graphics g) {
        String background = getParameter("background");
        String foreground = getParameter("foreground");
        String message = getParameter("message");
        setBackground(Color.decode(background));
        setForeground(Color.decode(foreground));
        Font font = getFont();
        FontMetrics fm = getFontMetrics(font);
        Dimension d = getSize();
        int x = (d.width - fm.stringWidth(message)) / 2;
        int y = d.height / 2;
        g.drawString(message, x, y);
    }
}
```



The AppletContext Interface

AppletContext Interface

```
Applet getApplet(String appName)
Enumeration getApplets()
AudioClip getAudioClip(URL url)
Image getImage(URL url)
void showDocument(URL url)
void showDocument(URL url, String target)
void showStatus(String str)
```

```
import java.applet.*;
import java.awt.*;
import java.net.*;
/*
<applet code="ShowDocument" width=200
height=50>
</applet>
*/

public class ShowDocument extends Applet {

    public void init() {
        AppletContext ac = getAppletContext();
        try {
            URL url = new URL("http://www.osborne.com");
            ac.showDocument(url, "frame2");
        }
        catch(Exception e) {
            showStatus("Exception: " + e);
        }
    }

    public void paint(Graphics g) {
        g.drawString("ShowDocument Applet", 10, 25);
    }
}
```



Using Images

getImage() Methods

Image getImage(URL url)
Image getImage(URL base, String fileName)

drawImage() Methods

abstract boolean drawImage(Image img, int x, int y, ImageObserver io)

```
import java.applet.*;
import java.awt.*;
/*
<applet code="DrawImage" width=280 height=280>
<param name="file" value="kids2.jpg">
</applet>
*/

public class DrawImage extends Applet {
    Image image;

    public void init() {
        image = getImage(getDocumentBase(),
            getParameter("file"));
    }

    public void paint(Graphics g) {
        g.drawImage(image, 0, 0, this);
    }
}
```



Using Threads

update() and repaint() Methods

repaint() : request an update of the applet display

update() : clears the applet display with the background color and then invokes the paint() method

```
public class Counter extends Applet
implements Runnable {
    int counter;
    Thread t;

    public void init() {

        // Initialize counter
        counter = 0;

        // Start thread
        t = new Thread(this);
        t.start();
    }
}
```

```
public void run() {
    try {
        while(true) {
            // Request a repaint
            repaint();
            //Sleep before displaying next count
            Thread.sleep(1000);
            //Increment counter
            ++counter;
        }
    }
    catch(Exception e) { }
}

public void paint(Graphics g) {

    // Set Font
    g.setFont(new Font("Serif", Font.BOLD, 36));

    // Get font metrics
    FontMetrics fm = g.getFontMetrics();

    // Display counter
    String str = "" + counter;
    Dimension d = getSize();
    int x = d.width / 2 - fm.stringWidth(str) / 2;
    g.drawString(str, x, d.height / 2);
}
}
```



Double Buffering (No Double Buffering)

Double Buffering can be used to avoid display “flicker” in applets.

```
import java.applet.*;
import java.awt.*;
/*
<applet code="NoDoubleBuffer" width=300
height=100>
</applet>
*/

public class NoDoubleBuffer extends Applet
implements Runnable {
    int x = 0;
    Thread t;

    public void init() {

        // Start thread
        t = new Thread(this);
        t.start();
    }

    public void run() {
        try {
            while(true) {
```

```
                // Request a repaint
                repaint();

                //Sleep before update
                Thread.sleep(100);
            }
        } catch(Exception e) {
        }
    }

    public void paint(Graphics g) {

        // Draw filled circle
        Dimension d = getSize();
        g.fillOval(x, d.height / 4, 50, 50);

        // Increment x
        x += 5;
        if (x + 50 > d.width)
            x = 0;
    }
}
```



Double Buffering (With Double Buffering)

```
import java.applet.*;
import java.awt.*;
/*
<applet code="DoubleBuffer" width=300 height=100>
</applet>
*/

public class DoubleBuffer extends Applet
implements Runnable {
    int x = 0;
    Thread t;
    Image buffer;
    Graphics bufferg;

    public void init() {

        // Start thread
        t = new Thread(this);
        t.start();

        // Create buffer
        Dimension d = getSize();
        buffer = createImage(d.width, d.height);
    }

    public void run() {
        try {
            while(true) {

                //Request a repaint
                repaint();
```

```
                // Sleep before update
                Thread.sleep(100);
            }
        } catch(Exception e) {
        }
    }

    public void update(Graphics g) {
        paint(g);
    }

    public void paint(Graphics g) {

        //Get graphics object for buffer
        if (bufferg == null)
            bufferg = buffer.getGraphics();

        //Draw to buffer
        Dimension d = getSize();
        bufferg.setColor(Color.white);
        bufferg.fillRect(0, 0, d.width, d.height);
        bufferg.setColor(Color.black);
        bufferg.fillOval(x, d.height / 4, 50, 50);

        //Update screen
        g.drawImage(buffer, 0, 0, this);

        //Increment x
        x += 5;
        if (x + 50 > d.width)
            x = 0;
    }
}
```